

SWETA GHOSH

AI-Driven Frontend Developer | Aspiring AI Engineer Building End-to-End AI Applications with React, Python, FastAPI Computer Vision

Jharkhand, India | swetaghosh665@gmail.com | +91 7484963888

Portfolio: <https://swetaghosh-portfolio.netlify.app/>

LinkedIn: <https://www.linkedin.com/in/swetaghosh05/>

Github: <https://github.com/swetaghosh180>

PROFESSIONAL SUMMARY

AI-Driven Frontend Developer and Aspiring AI Engineer with a strong foundation in modern frontend development and hands-on experience building end-to-end AI-powered applications. Proficient in React, Vue.js, JavaScript, Python, FastAPI, Scikit-learn, OpenCV, YOLOv8, EasyOCR, BLIP, and PyTorch, with practical experience integrating Machine Learning and Computer Vision models into responsive web applications. Completed a Frontend Developer Internship, where I developed scalable user interfaces, reusable components, and REST API integrations using Vue.js and Tailwind CSS. Passionate about solving real-world problems through intelligent software, continuously learning modern AI technologies, and building scalable, production-ready applications. Seeking opportunities to contribute and grow as an AI Engineer, Machine Learning Engineer, Computer Vision Engineer, Python Developer, or AI-Driven Frontend Developer.

TECHNICAL SKILLS

- **Programming Languages**
Python • JavaScript (ES6+) • SQL
 - **Frontend Development**
React.js • Vue.js • HTML5 • CSS3 • Tailwind CSS • Bootstrap • Responsive Design • Component-Based Architecture • REST API Integration
 - **Backend Development**
FastAPI • Flask • REST API Development
 - **Machine learning**
Scikit-learn • Regression • Classification • Decision Trees • Random Forest • KNN • Data Preprocessing • Feature Engineering • Model Evaluation • Model Deployment
 - **Computer Vision**
OpenCV • YOLOv8 • EasyOCR • BLIP • Object Detection • Optical Character Recognition (OCR) • Image Captioning • Image Processing • Edge Detection • Thresholding • Contours • Morphological Operations
 - **Deep learning**
PyTorch • CNNs • Neural Networks • Transfer Learning (Basic) • Transformers (Basic)
 - **Databases**
SQLite • PostgreSQL (Basic)
 - **Developer Tools**
Git • GitHub • VS Code • Jupyter Notebook • Google Colab • Postman
 - **AI Productivity Tools**
Amazon Q • ChatGPT • GitHub Copilot
-

FEATURED PROJECTS

- ❑ **VisionInspect AI – AI-Powered Image Analysis Platform** **Jul 2026 –Aug2026**
Tech Stack: Python • FastAPI • React • Tailwind CSS • OpenCV • YOLOv8 • EasyOCR • BLIP • SQLite

- Developed an end-to-end AI-powered image analysis platform capable of detecting objects, extracting text, and generating image captions.
- Integrated YOLOv8 for real-time object detection with confidence-based predictions.
- Implemented EasyOCR for multilingual Optical Character Recognition (OCR) from uploaded images.
- Built an automated image captioning pipeline using BLIP Transformer models.
- Generated structured PDF analysis reports containing detected objects, extracted text, and AI-generated summaries.
- Designed a responsive React frontend with image preview, upload workflow, loading states, and downloadable reports.
- Built REST APIs using FastAPI for seamless frontend-backend communication.
- Deployed the frontend using Netlify and configured the backend for local execution due to AI model memory requirements.

▣ **AI Plant Doctor**

(In Progress)

Tech Stack: Python • OpenCV • PyTorch • FastAPI • React

- Developing a computer vision-based application to detect plant diseases from leaf images using PyTorch and OpenCV.
- Integrating image preprocessing, deep learning models, and a FastAPI backend to provide disease predictions and treatment recommendations.

▣ **House Price Prediction System**

July 2026

Tech Stack: Python • Scikit-learn • Pandas • NumPy • Flask • HTML • CSS • JavaScript

- Developed an end-to-end machine learning application for predicting house prices using regression algorithms.
- Performed data cleaning, preprocessing, feature engineering, and exploratory data analysis.
- Trained and evaluated Linear Regression and Random Forest Regression models.
- Compared model performance using MAE, RMSE, and R² Score.
- Built a Flask-based web application allowing users to predict house prices through an interactive interface.

❖ **FRONTEND PROJECTS**

▪ **Developer Portfolio Website**

Jan 2026 – Mar 2026

Tech Stack: React • Tailwind CSS • JavaScript

- Designed and developed a responsive portfolio website using React and Tailwind CSS.
- Built reusable components following modern frontend development practices.
- Implemented responsive layouts optimized for desktop, tablet, and mobile devices.
- Showcased technical projects, certifications, and professional experience through an interactive interface.

▪ **eBhakti – Digital Scripture Library**

Mar 2026 – Jun 2026

Tech Stack: Vue.js • Vue Router • Tailwind CSS

- Built a multilingual Single Page Application with reusable components and responsive layouts.

▪ **Sweta's Kitchen Recipe**

Sept 2025 – Oct 2025

Tech Stack: HTML • CSS • JavaScript

- Developed a multilingual responsive recipe website featuring interactive UI and category-based navigation.

- Aurelia Luxe| Luxury Jewelry E-Commerce Website 2025

Oct 2025 – Dec

Tech Stack: HTML • CSS • JavaScript

- Designed and developed a premium responsive e-commerce frontend with reusable components and modern UI.

INTERNSHIP EXPERIENCE

❖ Roadlyft Rideshare Pvt. Ltd.

Nov 2025 – Jan 2026

Frontend Developer Intern

- Developed responsive and user-friendly web interfaces using Vue.js, Tailwind CSS, HTML5, and JavaScript.
- Built reusable UI components and structured layouts to support scalable and maintainable frontend architecture.
- Integrated RESTful APIs with frontend components to display dynamic financial data.
- Applied UI/UX best practices and accessibility standards to improve usability and overall user experience.
- Collaborated with designers and backend developers to implement features and optimize interface performance.
- Debugged and resolved frontend issues while improving application performance and responsiveness.

CERTIFICATIONS

- TATA GenAI Powered Data Analytics Certificate (Aug 2026)
- Python Certification – FreeCodeCamp (Inprogress)

EDUCATION

- Bachelor of Computer Applications (BCA)

Netaji Subhas University, Jamshedpur, Jharkhand (Aug 2023 – Aug 2026)

TARGET ROLES

- Artificial Engineer (Aspiring)
- Machine Learning Engineer
- Computer Vision Engineer
- AI-Driven Frontend Developer
- Python Developer
- Frontend Developer

PROFESSIONAL HIGHLIGHTS

- Strong foundation in modern frontend development with React and Vue.js.
- Transitioning into AI through hands-on projects in Machine Learning and Computer Vision.
- Experienced in building complete AI-powered web applications using React, FastAPI, and Python.
- Passionate about solving real-world problems through scalable, user-focused software solutions.
- Committed to continuous learning, clean code, and modern software engineering practices.